Practice Question

NLPP

Type-1: WITHOUT CONSTRAINT

Find maximum or minimum of the following functions

1. $z = x_1{}^2 + x_2{}^2 + x_3{}^2 - 6x_1 - 8x_2 - 10x_3$

2. $z = x_1 + 2x_3 + x_2x_3 - x_1{}^2 - x_2{}^2 - x_3{}^2$

3. $z = x_1{}^2 + x_2{}^2 + x_3{}^2 - 6x_1 - 10x_2 - 14x_3 + 103$

4. $z = 2x_1 + x_3 + 3x_2x_3 - x_1{}^2 - 3x_2{}^2 - 3x_3{}^2 + 17$

5. $z = x_1{}^2 + x_2{}^2 + x_3{}^2 - 8x_1 - 10x_2 - 12x_3 + 100$

6. $z = 2x_1 + 6x_3 + 9x_2x_3 - 4x_1{}^2 - 9x_2{}^2 - 9x_3{}^2$

TYPE-2: LAGRANGE'S METHOD

Using the method of Lagrange's multipliers, solve the following NLPP

1. Optimize $z = 2x_1{}^2 + 4x_2{}^2$ Subject to $x_1 + 3x_2 = 7, x_1, x_2 \geq 0$

2. Optimize $z = 2x_1{}^2 + 2x_2{}^2 + 2x_3{}^2 - 24x_1 - 8x_2 - 12x_3 + 260$

Subject to $x_1 + x_2 + x_3 = 11, x_1, x_2, x_3 \geq 0$

3. Optimize $z = 2x_1{}^2 + x_2{}^2 + 3x_3{}^2 + 10x_1 + 8x_2 + 6x_3 - 100$ Subject to $x_1 + x_2 + x_3 = 20, x_1, x_2, x_3 \geq 0$

4. Optimize $z = 4x_1 + 8x_2 - x_1{}^2 - x_2{}^2$ Subject to $x_1 + x_2 = 2, x_1,\ x_2 \geq 0$

TYPE-3 KUHN TUCKER CONDITIONS

Use the Kuhn-Tucker conditions to solve the following NLPP

1. Maximize $z = 4x_1 - x_1{}^3 + 2x_2$ Subject to $x_1 + x_2 \leq 1, x_1, x_2 \geq 0$

2. Minimize $z = -10x_1 - 4x_2 + 2x_1{}^2 + x_2{}^2$ Subject to $2x_1 + x_2 \leq 5, x_1, x_2 \geq 0$

3. Maximize $z = 2x_1{}^2 - 7x_2{}^2 + 12x_1x_2$ Subject to $2x_1 + 5x_2 \leq 98, x_1, x_2 \geq 0$

4. Maximize $z = 8x_1 + 10x_2 - x_1{}^2 - x_2{}^2$ Subject to $3x_1 + 2x_2 \leq 6, x_1, x_2 \geq 0$

5. Maximize $z = 16x_1 + 6x_2 - 2x_1{}^2 - x_2{}^2 - 17$ Subject to $2x_1 + x_2 \leq 8, x_1, x_2 \geq 0$